Summary in Graph

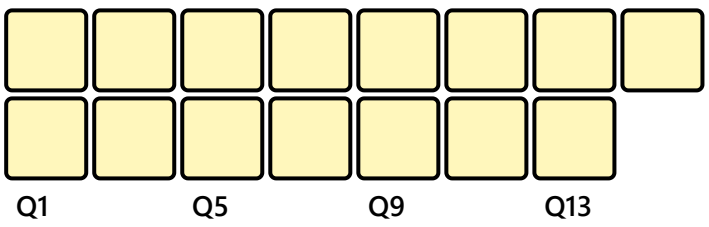
Exam Summary (GO Classes CS Test Series 2026 | Theory of Computation | Topic Wise Test 3)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	15 (5 + 10)
Total Marks:	25 (5 + 20)
Exam Duration:	45 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (116) out of 1155 users

EXAM RESPONSE EXAM STATS FEEDBACK

Technical

Q #1 Multiple Choice Type Award: 1 Penalty: 0.33 Theory of Computation

Consider the following language L with the alphabet $\{0, 1\}$. The empty string is not in L. If the string is nonempty and has even length, then it is in L. If the string has odd length, then it is in L if and only if it is a palindrome.

Which of the following best describes L?

A. Regular

B. DCFL but not regular
C. CFL but not DCFL
D. Not CFL

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Theory of Computation

Consider the following languages :

i. $\mathrm{L}_1 = \{ww \mid w \in \{a\}^*\}$
ii. $\mathrm{L}_2 = \{ww \mid w \in \{a, b\}^*\}$

Which of the above languages is Non-regular But CFL?

A. Only L_1
B. Only L_2
C. Both
D. None

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Theory of Computation

Consider the following languages :

1. The language of the CFG with productions $\mathrm{S} \rightarrow \mathrm{AS} \mid \mathrm{SB} \mid \epsilon$, $\mathrm{A} \rightarrow 0$, and $\mathrm{B} \rightarrow 1$
2. The language of regular expression 0^*1^*

Which of the following is true ?

A. 1 is proper subset of 2
B. 2 is proper subset of 1
C. $1 = 2$
D. Neither 1 is subset of 2, nor 2 is subset of 1.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Select Type | Award: 1 | Penalty: 0 | Theory of Computation

Which of the following languages are CFL But Not Regular?

A. $\mathrm{L} = \left\{a^{2n}b^{3n} \mid \mathrm{n} > 0\right\}$
B. $\mathrm{L} = \left\{a^{2n}b^{3m} \mid \mathrm{m}, \mathrm{n} > 0\right\}$
C. $\mathrm{L} = \left\{a^{2n}b^{3} \mid \mathrm{n} > 0\right\}$
D. $\mathrm{L} = \left\{a^{2n}b^{n+m} \mid \mathrm{n}, \mathrm{m} > 0\right\}$

Your Answer: | Correct Answer: A;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Compiler Design

A grammar is said to be "useless" if and only if it produces no terminal strings. If S is the start symbol, which of the following grammars is (are) "useless"?

I. $\begin{array}{l} \mathrm{S} \rightarrow \mathrm{AB}|\mathrm{AS} \\ \mathrm{A} \rightarrow \mathrm{B} \mid a \\ \mathrm{B} \rightarrow \mathrm{A} \mid b \end{array}$

II. $\begin{array}{l} \mathrm{S} \rightarrow \mathrm{SA} \mid \mathrm{AS}|\mathrm{SB} \\ \mathrm{A} \rightarrow a \\ \mathrm{B} \rightarrow a \mid b \end{array}$

III. $\begin{array}{l} \mathrm{S} \rightarrow \epsilon|\mathrm{A} \\ \mathrm{A} \rightarrow \mathrm{B} \\ \mathrm{B} \rightarrow \mathrm{C} \\ \mathrm{C} \rightarrow a \end{array}$

A. None
B. Only II
C. Only III
D. Only II, III

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Select Type | Award: 2 | Penalty: 0 | Theory of Computation

Consider the following Grammar:

$$\mathrm{S} \rightarrow rl \mid r\mathrm{S}l \mid \mathrm{SS}$$

Which of the following strings can be generated by this grammar?

A. $rrllrl$
B. $rllrrrll$
C. $rrlrl\ rllrl\ rrl$
D. $rrrll\ lrlr\ rll$

Your Answer: | Correct Answer: A;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Select Type | Award: 2 | Penalty: 0 | Compiler Design

Which of the following statements are true?

A. The grammar, $\mathrm{G} : \mathrm{E} \rightarrow \mathrm{E} + \mathrm{E} \mid \mathrm{E} * \mathrm{E} \mid x$ is ambiguous.
B. A regular grammar can never be ambiguous.
C. A grammar is ambiguous if there is some string w such that $w's$ right-most derivation differs from its left-most derivation.
D. A grammar is ambiguous if there is some string w such that w has two different parse trees.

Your Answer: | Correct Answer: A;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 Multiple Select Type | Award: 2 | Penalty: 0 | Theory of Computation

Let $\mathrm{L} = \{a^p \mid p \text{ is prime}\}$ be a language. Which of the following are true?

A. L is Not context free language.
B. L^* is Not Context free language.
C. Complement of L^* is regular.
D. Complement of L is CFL.

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 02sec | Discuss

Q #9 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Theory of Computation

A context-free grammar has a set of terminals $\{0, 1, -\}$, a set of non-terminals $\{\mathrm{N}, \mathrm{P}\}$, where N is the start symbol, and productions given by the following .

$$\mathrm{N} \rightarrow 0 \mid \mathrm{P} \mid - \mathrm{P}$$
$$\mathrm{P} \rightarrow 1 \mid \mathrm{P}\ 0 \mid \mathrm{P}\ 1$$

What can be said about the language generated by this grammar?

A. It is regular.
B. It is Non-regular But DCFL.
C. It is non-DCFL but CFL.
D. It is Non-CFL.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 Multiple Select Type | Award: 2 | Penalty: 0 | Theory of Computation

Which of the following languages are not context free?

A. $\mathrm{L}_1 = \left\{0^n 1^{2n} \mid n \geq 0\right\}$
B. $\mathrm{L}_2 = \left\{w \in \{0,1\}^* \mid w \neq w^{\mathrm{R}}\right\}$.
C. $\mathrm{L}_3 = \{www \mid w \in \{0,1\}^*\}$.
D. $\mathrm{L}_4 = \{x/y/z \mid x, y, z \in \{0,1\}^*$ are binary numbers such that $x + y = z\}$. The alphabet for this language is $\{0, 1, /\}$. For example, $10/10/100 \in \mathrm{L}_3$ and $11/1/001 \notin \mathrm{L}_3$.

Your Answer: | Correct Answer: C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Theory of Computation

Every day, a weather station records whether the day was sunny (S), cloudy (C) or rainy (R). A sequence of records over several days is a string in $\{\mathrm{S}, \mathrm{C}, \mathrm{R}\}^*$. We call a sequence of records GO-like if:

- There are never more than three consecutive rainy days, and
- There are never more than five consecutive non-sunny days.

The language of GO-like sequences is

A. Regular
B. DCFL but Not regular
C. CFL but not DCFL

D. Not CFL

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Theory of Computation

Consider the following languages:

1. The language (accepted by final state) of $\mathrm{PDA}(\{p,q\},\{0,1\},\{\mathrm{X},\mathrm{Z}\},\delta,q,\mathrm{Z},\{p\})$ with δ defined by the rules $\delta(q,0,\mathrm{Z})=\{(q,\mathrm{X}\ \mathrm{Z})\},\delta(q,0,X)=\{(q,\mathrm{X}\ \mathrm{X})\},\delta(q,1,\mathrm{X})=\{(p,\epsilon)\}$, and $\delta(p,1,\mathrm{X})=\{(p,\epsilon)\}$.
2. The language of the CFG with productions $\mathrm{S}\rightarrow 0\ \mathrm{S}\ 1 \mid 0\ \mathrm{S} \mid \epsilon$.

Which of the following is true ?

A. 1 is proper subset of 2
B. 2 is proper subset of 1
C. $1=2$
D. Neither 1 is subset of 2, nor 2 is subset of 1

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Select Type | Award: 2 | Penalty: 0 | Theory of Computation

Which of the following languages is/are Context Free(CFL)?

A. $\{w\#w' \mid w,w' \in \{a,b\}^* \text{ and } w \neq w'\}$.
B. $\{0^n1^m \mid n \leq m \leq 2n\}$.
C. $a^*b^*c^* - \{a^nb^nc^n \mid n \geq 0\}$.
D. $\{w \in \{0,1,2\}^* \mid \#_0(w)=\#_1(w)=\#_2(w)\}$

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Select Type | Award: 2 | Penalty: 0 | Theory of Computation

An infinite language is $\mathrm{REG\text{-}immune}$ if it contains no infinite subset that is a regular language.

Which of the following are $\mathrm{REG\text{-}immune}$ languages?

A. $\{a^nb^n \mid n \geq 0\}$
B. $\{ww \mid w \in (a+b)^*\}$
C. $\{w\#w \mid w \in (a+b)^*\}$
D. $\{a^p \mid p \text{ is prime}\}$

Your Answer: | Correct Answer: A;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Numerical Type | Award: 2 | Penalty: 0 | Compiler Design

$$\begin{aligned}\mathrm{S} &\rightarrow \mathrm{A0B}\\ \mathrm{A} &\rightarrow \mathrm{BB} \mid 0\\ \mathrm{B} &\rightarrow \mathrm{AA} \mid 1\end{aligned}$$

What is the number of terminal strings of length 5 generated by the context-free grammar shown above?

Your Answer: | **Correct Answer: 5** | **Not Attempted** | **Time taken: 00min 00sec** | **Discuss**

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